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Funciones medibles de variables aleatorias independientes

Corolario 1. *Sea $n \in \mathbb{N}$ y $\forall i \in \mathbb{N} : m(i) \in \mathbb{N}$ y sean*

(i) $\{X_{i,j}\}_{i,j=1}^{n,m(i)}$ *son variables aleatorias independientes*

(ii) $\forall i \in \mathbb{N}_n : f_i \colon \mathbb{R}^{m(i)} \longrightarrow \mathbb{R}$ *medible.*

$$\implies \left\{f_i(X_{i,1}, \ldots, X_{i,m(i)})\right\}_{i=1}^{n} \text{ son independientes.}$$

***Demostración*:** Vamos a dividir la demostración en pasos:

1. Definimos $\forall i \in \mathbb{N} : X_i \colon \Omega \longrightarrow \mathbb{R}^{m(i)}$ dada por $X_i(\omega) = \left(X_{i,1}(\omega), \ldots, X_{i,m(i)}(\omega)\right)$. Por el apartado `ejer-var-aleatorias-prod-suma/??` del `ejer-var-aleatorias-prod-suma/??`, X_i es un vector aleatorio. Veamos que $\{X_i\}_{i=1}^{n}$ son independientes. Sean $\forall i \in \mathbb{N} : E_i \in \mathcal{B}\left(\mathbb{R}^{m(i)}\right)$ queremos ver que
$$\mathbb{P}\left(\bigcap_{i=1}^{n} X_i^{-1}(E_i)\right) = \prod_{i=1}^{n} \mathbb{P}\left(X_i^{-1}(E_i)\right).$$
Basta probarlo para los E_i de la forma $E_i = E_{i,1} \times \cdots \times E_{i,m(i)}$ con $E_{i,j} \in \mathcal{B}(\mathbb{R})$, ya que $\mathcal{B}\left(\mathbb{R}^{m(i)}\right)$ está generada por estos conjuntos. Entonces
$$\forall i \in \mathbb{N}_n : X_i^{-1}(E_i) = \left\{\omega \in \Omega : \forall j \in \mathbb{N}_{m(i)} : X_{i,j}(\omega) \in E_{i,j}\right\} = \bigcap_{j=1}^{m(i)} X_{i,j}^{-1}(E_{i,j}).$$
$$\implies \mathbb{P}\left(\bigcap_{i=1}^{n} X_i^{-1}(E_i)\right) = \mathbb{P}\left(\bigcap_{i=1}^{n}\bigcap_{j=1}^{m(i)} X_{i,j}^{-1}(E_{i,j})\right) \overset{\substack{\text{indep}\\ \downarrow}}{=} \prod_{i\in\mathbb{N}_n}\prod_{j=1}^{m(i)} \mathbb{P}\left(X_{i,j}^{-1}(E_{i,j})\right)$$
$$= \prod_{i=1}^{n}\prod_{i\in\mathbb{N}_n} \mathbb{P}\left(\bigcap_{j=1}^{m(i)} X_{i,j}^{-1}(E_{i,j})\right) = \prod_{i=1}^{n} \mathbb{P}\left(X_i^{-1}(E_i)\right).$$

2. Observamos que $\forall i \in \mathbb{N}_n : f_i\left(X_{i,1}, \ldots, X_{i,m(i)}\right) = f_i \circ X_i$, entonces si consideramos

$$\forall i \in \mathbb{N}_n : E_i \in \mathcal{B}(\mathbb{R}),\ (f_i \circ X_i)^{-1}(E_i) = X_i^{-1}\left(f_i^{-1}(E_i)\right) \text{ donde } f_i^{-1}(E_i) \in \mathcal{B}\left(\mathbb{R}^{m(i)}\right).$$

$$\implies \mathbb{P}\left(\bigcap_{i\in\mathbb{N}_n} (f_i \circ X_i)^{-1}(E_i)\right) = \mathbb{P}\left(\bigcap_{i\in\mathbb{N}_n} X_i^{-1}\left(f_i^{-1}(E_i)\right)\right) \overset{??}{\underset{\downarrow}{=}} \prod_{i\in\mathbb{N}_n} \mathbb{P}\left(X_i^{-1}\left(f_i^{-1}(E_i)\right)\right).$$

Por tanto, $\{f_i \circ X_i\}_{i=1}^{n}$ son variables aleatorias independientes. ■

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